

Cancellation of Self-sensing due to Ego-motion of Vibration-based Mechanosensors

1. Introduction

Tactile sensing is an important capability in both biological organisms and autonomous robots. Many insects use antennae as active tactile sensors to detect obstacles, recognize surfaces, and navigate through their environments. Inspired by these biological systems, we can develop vibration-based robotic antennae using flexible probes and accelerometers. These systems are especially useful in low-light or visually difficult environments where cameras are less effective. However, when the robot moves, vibrations it generates create false sensor readings. This effect is called self-sensing or ego-motion disturbance. The main objective of this project is to separate true contact signals from self-generated vibrations using signal processing and machine learning techniques.

Three primary strategies have been recognized in the literature to tackle this issue: (1) Echo State Networks (ESN) utilized for forecasting and mitigating self-generated vibrations via a forward model; (2) adaptive filtering methods like Kalman filtering that provide efficient noise reduction; and (3) deep learning techniques including CNNs and RNNs which attain improved accuracy at increased computational expense. The methods discussed thoroughly in the subsequent sections establish the theoretical basis for the cancellation algorithm created in this project.

2. Biological Inspiration of Robotic Antennae

According to Staudacher et al. (2005), insect antennae are highly specialized tactile sensing organs equipped with mechanoreceptors and proprioceptors. These structures allow insects to actively explore nearby objects, identify obstacles, and understand environmental properties. The review explains that insects such as stick insects, crickets, cockroaches, locusts, and honeybees use antennae for active sensing during locomotion. The antennae can move in different directions using joints and muscles, allowing wide spatial coverage.

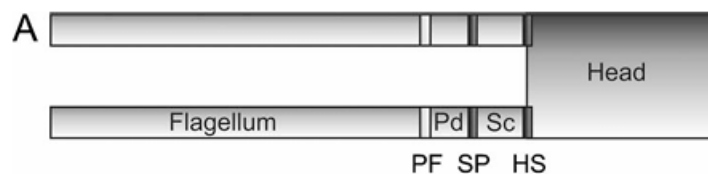


Figure 1: Schematic top view of an insect head. The antennae consist of three functional segments: the scape (Sc), pedicel (Pd), and an annulated flagellum.

The antenna typically consists of three main parts: the scape, pedicel, and flagellum. The flagellum acts as the primary sensing region. When the antenna touches an object, mechanical vibrations are transferred through the structure and can be measured using mechanoreceptors. The same principle is used in artificial robotic antenna systems, where an accelerometer is mounted at the tip of a flexible probe.

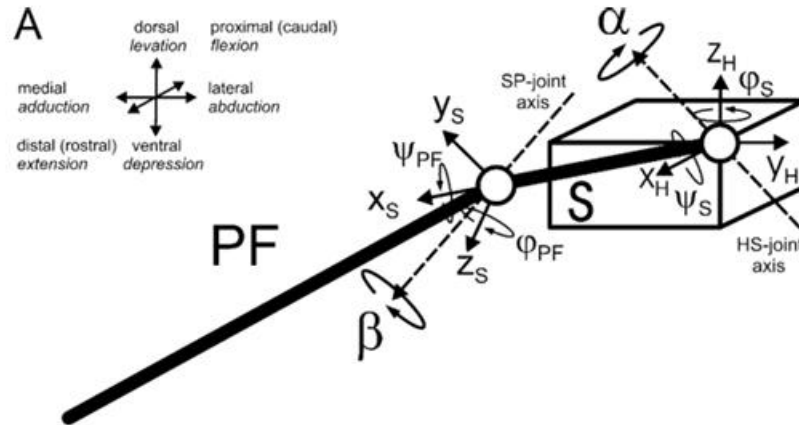


Figure 2: Kinematic Model of Insect Antenna Movement

The paper also explains that insects can distinguish between self-generated motion and external touch by combining motor information with sensory feedback. This concept directly inspired the development of forward models and bio-inspired algorithms used in modern robotic tactile systems.

Phase-Dependent Sensing

Dürr et al. (2003) showed that insect antennae serve as dynamic tactile sensors that continually shift to collect spatial data. Significantly, their study demonstrated that antennal sensitivity varies with phase, indicating the antenna's reaction to touch alters depending on the phase of its movement cycle at that moment. This is closely related to the ego-motion issue in this project: the motion state of the sensor influences its interpretation of incoming signals, and any cancellation algorithm needs to consider this dynamic connection between movement phase and sensory output.

The Efference Copy Principal

A fundamental biological process for cancelling self-motion is the efference copy principle, proposed by Miall & Wolpert (1996) regarding physiological motor control. When a biological system creates a motor command to move a limb or appendage, it concurrently transmits a replica of that motor command (an efference copy) to the

sensory processing centers. The sensory system employs this anticipated self-motion signal to eliminate the expected self-created component from incoming sensory information, resulting in just the externally-induced signal. This represents the biological counterpart of the forward model technique presented in Paper [1] (Section 3), and it directly supports the application of motor command signals as inputs to the ESN cancellation model within this project.

Animal Whisker-based Sensing

In addition to insects, mammals also serve as significant biological models. Prescott et al. (2011) examined how rats use rhythmic whisking actions, their whiskers oscillate rapidly in the environment to capture information about object distance, shape, and surface texture via mechanical bending. Harbor seals utilize comparable whisker-based sensing to monitor water currents and moving items in environments with low visibility. These systems encounter the same self-sensing issue: while whisking quickly, the animal needs to distinguish between bending caused by contact and the self-initiated movement of the whisker sweep. The neural processes enabling this separation in rodents, utilizing proprioceptive feedback and previously learned movement models, closely resemble the adaptive filtering methods discussed in Section 3.



Figure 3: Block Diagram of the Process

3. Review of Main Research Papers

Paper [1]– A Forward Model for an Active Tactile sensor using Echo State Networks

This study introduced a forward model based on Echo State Networks (ESN) to predict self-generated vibration signals in a robotic antenna system. The robotic antenna consisted of a flexible plastic tube with an accelerometer mounted at the tip. The antenna was attached to a mobile robot using a pan-tilt mechanism. Inputs to the ESN included robot acceleration data and servo motor position signals. The study showed that ESNs can successfully predict sensor readings during robot movement, thereby enabling the identification of external touch events.

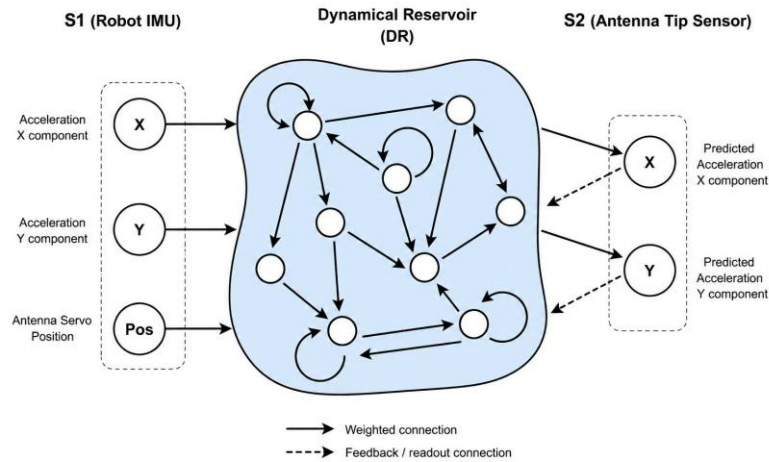


Figure 3: Echo State Network showing the inputs, S1 and antennal servo position (Pos), Dynamical reservoir (DR), and the output S2 (signal to be predicted). X and Y represent the two components of the acceleration vector in each sensor.

Paper [2] – Prediction of Self-induced Mechanoreceptive Sensor Readings in an Insect-inspired Active Tactile Sensing System

This paper expanded the previous work by introducing two antennae with multiple motion patterns. Experiments included straight robot motion, curved motion, and rhythmic disturbances created using spiked wheels. The researchers used signal rectification and low-pass filtering to improve touch event detection. The study demonstrated that forward models can significantly reduce false detections caused by ego-motion.

Paper [3] – Distance Estimation and Material Classification of a Compliant Tactile Sensor Using Vibration Modes and Support Vector Machine

This research focused on using vibration analysis for estimating contact distance and identifying material types. FFT analysis, Artificial Neural Networks (ANN), and Support Vector Machines (SVM) were compared. The researchers observed that vibration characteristics change depending on the contact location along the antenna. SVM-based approaches provided improved performance for classification tasks.

Paper [4] – Forward Models and the Efference Copy Principle

Miall and Wolpert (1996) presented the idea of forward models in the context of physiological motor control. They suggested that the brain holds an internal representation of the body that forecasts the sensory outcomes of motor commands. A duplicate of the motor command, the efference copy, is input into this internal model to produce a predicted sensory state. By deducting this forecast from the real sensory input, the system separates signals generated by external occurrences instead of self-movement. This biological principle directly underlies the ESN-based forward model presented in Paper [1] and supports using servo motor position and robot acceleration as inputs for the cancellation model. For this project, the Arduino and IMU data streams act as the efference copy inputs to the predicting algorithm.

Paper [5] – Adaptive Filtering and Kalman-Based Approaches (Russell & Wijaya, 2003)

Prior to the dominance of neural network methods, Kalman filters and Fourier-based filtering were the main techniques for reducing ego-motion artifacts in systems using whiskers and antennas. Russell & Wijaya (2003) utilized these methods to eliminate self-sensing noise in robotic whisker systems, accomplishing about 70–80% reduction of artifacts in organized settings. Although it is computationally efficient and simple to implement, Kalman filtering presumes that motion artifacts display linear and predictable behaviors. In real applications, robot motion on irregular surfaces generates non-linear, non-stationary vibration patterns that contradict this assumption. This constraint directly encourages the application of ESN-based methods in this project, since reservoir computing can represent non-linear dynamic systems without the need for explicit mathematical modeling of the noise origin.

Paper [6] – Deep Learning for Motion Artifact Suppression

Recent studies have utilized deep learning, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), to address the challenge of distinguishing self-generated noise from tactile signals. CNN-based methods have reached accuracy levels of up to 95% in texture classification while also eliminating motion artifacts. Nonetheless, these approaches demand extensive labeled training

datasets and significant computational power, which complicates their implementation on low-power embedded systems like the Arduino Uno and Jetson Nano utilized in this project. Deep learning continues to be a promising option for future versions of the system after adequate training data is gathered; however, the current project emphasizes ESN-based methods as a more feasible initial approach for embedded real-time cancellation.

4. Comparison of Reviewed Studies

Research Paper	Main Technique	Sensors Used	Main Objective	Advantages	Limitations
[1]	Echo State Network	Accelerometer + Servo Position	Predict self-generated vibrations	Good prediction capability	Requires parameter tuning
[2]	ESN + Filtering	IMU + Accelerometers	Detect true touch events	Works under multiple motions	Complex preprocessing required
[3]	FFT + ANN + SVM	MPU6050 Accelerometer	Distance & material estimation	High classification accuracy	Reduced accuracy near antenna tip

5. Comparison of Cancellation Techniques

To determine the most suitable ego-motion cancellation method for this project, the three main approaches identified in the literature are compared below across key performance and practical criteria.

Technique	Mechanism	Performance	Limitation	Relevance to This Project
Kalman Filtering	Linear state estimation; real-time, low computational cost	70–80% artifact reduction in structured environments	Fails with non-linear vibrations (e.g. mobile robot on uneven terrain)	Limited robot ego-motion is non-linear and non-stationary
Echo State Network (ESN)	Reservoir computing; models non-linear dynamic noise	90% reduction in unstructured settings	Requires GPU for real-time; needs parameter tuning	Primary approach for this project. Handles non-linear ego-motion

Deep Learning (CNN/RNN)	Spatial feature extraction + temporal prediction	95% accuracy in texture classification	High compute cost large dataset required	Future direction
-------------------------	--	--	---	------------------

Based on this comparison, the ESN-based forward model approach is selected as the primary cancellation method for this project. It provides the best balance between cancellation performance and computational feasibility on the Jetson Nano platform. Kalman filtering will be considered as a lightweight baseline for comparison during evaluation. Deep learning methods are noted as a future direction once sufficient labelled training data is available from the experimental phase.

6. Challenges and Open Problems

Although existing studies successfully reduced self-generated sensor disturbances, several challenges still remain. Real-time processing is computationally demanding, especially when multiple sensors are used. The performance of machine learning models depends heavily on training conditions and parameter selection. Furthermore, many systems are tested only in controlled laboratory environments. More research is needed to improve robustness under real-world conditions such as uneven terrain, continuous robot motion, and environmental noise.

Another important challenge is adaptive learning. Most current systems use pre-trained models and fixed thresholds. Future systems should automatically adapt to changing conditions and robot dynamics. Lightweight embedded implementations are also necessary for practical mobile robotic applications.

Adaptive Learning Gap: Most current cancellation systems utilize models that are trained offline and implemented with static parameters. Under real-world circumstances, the vibration traits of a robot alter with age, varying loads, or when traversing various terrains. A system that fails to adjust to these changes will decline in performance as time progresses. This project recognizes this gap and aims to investigate if the ESN model can be incrementally retrained or fine-tuned with newly acquired data, without the need for a complete retraining cycle.

Real-World vs. Laboratory Gap:

The majority of evaluated systems were tested solely in controlled laboratory settings with flat surfaces using restricted, pre-defined motion patterns. This initiative directly tackles this issue by attaching the antenna to a mobile robot platform featuring a holonomic 4WD drive system, allowing diverse and realistic ego-motion profiles, such as lateral movement,

rotation in place, and combined directional motions, which are unachievable with standard differential-drive platforms used in previous research.

7. Research Gap and Project Positioning

The reviewed literature demonstrates strong progress in bio-inspired tactile sensing and ego-motion cancellation. However, three key gaps remain unresolved:

1. **Real-time embedded processing:** Most high-performance cancellation methods (deep learning, multi-sensor ESN) require computational resources beyond what is available on low-power mobile robot platforms. There is a need for lightweight algorithms that can run in real time on embedded hardware such as the Jetson Nano.
2. **Adaptation to changing robot dynamics:** Pre-trained, fixed-parameter models cannot reliably handle changing robot conditions. Systems that adapt online to new motion profiles and payload conditions are needed but remain underdeveloped.
3. **Testing beyond controlled environments:** The majority of prior work was validated on flat surfaces with pre-programmed motion. Real-world deployment on a holonomic mobile robot with varied motion profiles has not been widely studied.

This project addresses gaps (1) and (3) by implementing an ESN-based bio-inspired cancellation algorithm on a Jetson Nano-equipped 4WD holonomic mobile robot, and validating it under diverse ego-motion scenarios. Gap (2) is identified as a future research direction.

8. Conclusion

The reviewed literature shows that bio-inspired tactile sensing systems are highly effective for robotic obstacle detection and environmental interaction. Researchers have successfully used vibration analysis, signal processing, and machine learning to reduce self-sensing caused by robot motion. Echo State Networks, FFT analysis, ANN, and SVM methods have shown promising results for predicting and classifying tactile signals. These studies provide a strong foundation for the current project on the cancellation of self-sensing due to the ego-motion of vibration-based mechanosensors.

The algorithm design choices made in this project are influenced by the biological efference copy principle, adaptive filtering techniques, and ESN-based forward models. The recognized research gaps, especially concerning embedded real-time processing and testing of holonomic platforms, validate the originality and importance of this study in relation to the present condition of the discipline.

9. References

- [1] Harischandra, N., & Dürr, V. (2012). “A Forward Model for an Active Tactile Sensor Using Echo State Networks”.
- [2] Harischandra, N., & Dürr, V. (2013). “Prediction of Self-induced Mechanoreceptive Sensor Readings in an Insect-inspired Active Tactile Sensing System”.
- [3] Gunasekara, S. R., et al. (2019). “Distance Estimation and Material Classification of a Compliant Tactile Sensor Using Vibration Modes and Support Vector Machine”.
- [4] Staudacher, E. M., Gebhardt, M., & Dürr, V. (2005). “Antennal Movements and Mechanoreception: Neurobiology of Active Tactile Sensors”.
- [5] T. J. Prescott, M. E. Diamond, and A. M. Wing, “Active touch sensing,” *Philosophical Transactions of the Royal Society B: Biological Sciences*, vol. 366, no. 1581, pp. 2989–2995, Oct. 2011.
- [6] Russell, R. A., & Wijaya, J. A. (2003). “Object Location and Recognition Using Whisker Sensors”.
- [7] Prescott, T. J., et al. (2011). *Active Touch Sensing in Mammals and Robots*. *Philosophical Transactions of the Royal Society B*.